

Principles of Microeconomics — Week 3, Day 1

In-Class Problem Set: Introduction to Game Theory

This problem set accompanies the Day 1 session. Numbers differ from the lecture examples. Solutions (instructor copy) are collected at the end; set `\solnfalse` in the preamble to produce a student version.

Exercise 1 — Reading three games

For each game the row player's payoff is listed first. Find every pure-strategy Nash equilibrium, note any dominant strategy, and say what *type* of game it is (Prisoner's Dilemma, coordination, or zero-sum).

Game A — two firms choose to Advertise (Adv) or Not.

	Adv	Not
Adv	(3, 3)	(6, 1)
Not	(1, 6)	(5, 5)

Game B — two firms adopt a common standard X or Y.

	X	Y
X	(2, 2)	(0, 0)
Y	(0, 0)	(3, 3)

Game C — a zero-sum game (entries in each cell sum to zero).

	C	D
Top	(2, -2)	(-1, 1)
Bottom	(-1, 1)	(1, -1)

- Solve Game A. Is there a dominant strategy? Is the equilibrium efficient? Which classic game is it?
- Solve Game B. How many pure equilibria are there, and which do the players prefer? What is the coordination problem?
- Solve Game C. Show there is no pure-strategy equilibrium.

Exercise 2 — Mixing, and a sequential game

Part A — Mixed strategy. Use Game C above. Let the row player play Top with probability p , and the column player play C with probability q .

- Choose q to make the row player indifferent between Top and Bottom.
- Choose p to make the column player indifferent between C and D. State the mixed-strategy Nash equilibrium.

Part B — Sequential game. A small firm (the **Entrant**) chooses to enter (In) or stay Out. If it enters, the **Incumbent** chooses to Fight or Accommodate. Payoffs are (Entrant, Incumbent):

Out $\rightarrow$ (0, 8); In then Accommodate $\rightarrow$ (3, 3); In then Fight $\rightarrow$ (-1, 1).

- (c) Draw the game tree and solve it by backward induction.
- (d) Is “Fight” a credible threat? What would the incumbent need in order to deter entry?

Reconvene: for each game state how many equilibria it has and of what type; for the tree, state the credible outcome.